

# Derrin Bright

## Bioinformatics | Biotechnology | ML in Biology

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### Education

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| <b>National University of Singapore (NUS)</b><br><i>Master of Science in Precision Health and Medicine, Yong Loo Lin School of Medicine</i> | Singapore<br>Aug 2026-Present            |
| <b>Vellore Institute of Technology (VIT)</b><br><i>Bachelor of Technology in Biotechnology</i>                                              | CGPA: 8.8    Vellore, India<br>2022-2026 |

### Technical Skills

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| <b>Computational Biology Tools</b> | BLAST   AlphaFold   AutoDock Vina   GROMACS   MGLTools   Bioconductor   OpenBabel   Cytoscape   Seurat   GSEA   FIMO   DESeq2   WGCNA   MSigDB   limma                       |
| <b>Data Analysis Tools</b>         | Neo4J   KNIME   NetworkAnalyst   TCGA/GDC                                                                                                                                    |
| <b>Programming Tools</b>           | R   Python                                                                                                                                                                   |
| <b>Machine Learning Tools</b>      | TensorFlow   Keras   NumPy   Pandas   Matplotlib   scikit-learn   XGBoost                                                                                                    |
| <b>Certificates</b>                | Machine Learning A-Z   Python for Data Science, AI & Development   Bulk RNA-Seq Data Analysis   Advanced Bioinformatics   Single-Cell RNA-Seq Data Analysis using R & Python |

### Research Experiences

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| <b>Project Intern at Nanyang Technological University, Singapore</b><br><ul style="list-style-type: none"><li>Developed a transcription factor activity inference pipeline on large-scale postmortem brain RNA-seq data to investigate gene regulatory changes across neurodegenerative disease progression.</li><li>Used VIPER to infer regulon activity from the RNA-seq counts, then modelled group differences with linear mixed-effects models, permutation testing and surrogate variable analysis to account for cohort-level confounders.</li><li>Supervised by Dr. Guillaume Thibault   Guillaume Thibault Lab</li></ul>                                                                                                                                                                                                                                                                                                                | Jan 2026 - June 2026    |
| <b>Research Intern at Boston University, USA</b><br><ul style="list-style-type: none"><li>Scanned Long Control Region sequences across 207 HPV types and identified 104 transcription factors with significantly different binding site densities between high-risk and low-risk types.</li><li>Integrated results with TCGA transcriptomics and public ChIP-seq data, finding over half of the high-risk-enriched TFs expressed in HPV-positive tumors and confirming LCR occupancy for seven of them.</li><li>Analyzed sequence conservation across HPV-16 and HPV-18 clinical isolates and found the identified binding sites were highly conserved across strains.</li><li>First author on the resulting paper in <i>Viruses</i> (2026).</li><li>Supervised by Dr. Juan Fuxman Bass</li></ul>                                                                                                                                                | Sept 2025 - April 2026  |
| <b>Research Intern at University of Liverpool, UK</b><br><ul style="list-style-type: none"><li>Built and released a catalogue of 30 genome-scale metabolic models for five immune cell types across CLL, Richter Transformation and healthy donors, reconstructed from a large single-cell bootstrap ensemble using ftlNIT and GECKO 3 enzyme constraints.</li><li>Compared the transformed state against both CLL and a matched healthy baseline to separate metabolic capabilities that were genuinely new from ones that were simply restored.</li><li>Ran flux balance analysis and flux variability analysis across the model catalogue to characterise metabolic capability and flux behaviour for each cell type and condition.</li><li>Supervised by Dr. Francesco Del Carratore</li></ul>                                                                                                                                               | Sept 2025 - August 2026 |
| <b>Research Intern at IIT Bombay, India</b><br><ul style="list-style-type: none"><li>Performed differential expression analysis and WGCNA across four autoimmune diseases (SLE, rheumatoid arthritis, type 1 diabetes, ankylosing spondylitis) to identify disease-associated co-expression modules and hub genes.</li><li>Followed up with Gene Set Enrichment Analysis and PPI network construction using NetworkAnalyst to uncover dysregulated pathways and functional gene relationships shared across the four diseases.</li><li>Built and evaluated classification models (Logistic Regression, SVM, XGBoost and a deep neural network) to distinguish disease from healthy states based on gene expression, and tested how well the models generalized across the four diseases.</li><li>Supervised by Dr. Prakriti Tayalia</li></ul>                                                                                                    | May 2025 - Sept 2026    |
| <b>Research Intern at Ashoka University, India</b><br><ul style="list-style-type: none"><li>Analyzed five assembled <i>E. coli</i> genomes for carbapenem and beta-lactam resistance using BLAST and RGI-CARD to identify mutations, and correlated genotype-phenotype relationships through AST report analysis.</li><li>Performed Multiple Sequence Alignment (MSA) of Penicillin-Binding Proteins (PBPs) from the five <i>E. coli</i> genomes against a reference wild-type sequence to detect conserved mutations, and conducted genomic analysis that led to the identification of a 12-base pair insertion in the PBP3 gene, contributing to antibiotic resistance mechanisms.</li><li>Automated genomic data processing by developing a custom script to systematically rename and organize FASTA files, creating a structured dataset for phylogenetic analysis using the PhaME tool.</li><li>Supervised by Dr. Shraddha Karve</li></ul> | Jan 2025 - Mar 2025     |
| <b>Biotech Team Lead at NovaLinks, India</b><br><ul style="list-style-type: none"><li>Led a project integrating miRNA analysis and histopathological imaging with machine learning to enhance lung cancer subtype classification and improve diagnostic precision.</li><li>Conducted research on miRNA biomarkers specific to lung cancer and provided biological insights to support the development of an accurate classification model for distinguishing adenocarcinoma from squamous cell carcinoma.</li></ul>                                                                                                                                                                                                                                                                                                                                                                                                                              | Feb 2025 - April 2025   |

## Publications

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**Comparative Analysis of Transcription Factor Binding Sites in the Long Control Region Across Human Papillomavirus Types** | *Viruses*, 2026, 18(6):646 (MDPI)

**Integrative transcriptomic analysis identifies predictive gene signatures across autoimmune diseases: implications for personalized therapeutic targeting** | *Under review at Journal of Autoimmunity (Elsevier)*

**HPV-Associated Transcriptional Landscape in Cervical and Head and Neck Cancers** | *In revision at Infectious Agents and Cancer (Springer Nature)*

**Beyond the HPV Imprint: Cervicovaginal Dysbiosis and Microbiome-Mediated Modulation of HPV Pathogenesis and Cervical Cancer Progression** | *Submitted to Scientifica (Hindawi)*

## Technical Projects

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**Computational Identification of Antimalarial Leads through Fingerprint Similarity Analysis** [GitHub Link](#)  
• Screened a library of antimalarial molecules against a query (chloroquine drug) by generating molecular fingerprints and computing Tanimoto similarity coefficients using KNIME and RDKit nodes.

- Extracted molecular data from the Protein Data Bank (PDB) and visualized pairwise similarity scores as a heatmap to identify high-similarity candidates for antimalarial drug development.

**In Silico Analysis of 3DTC/CEP-6331 Complex using Molecular Docking** [GitHub Link](#)  
• Implemented a molecular docking simulation using AutoDock Vina to model the interaction between the 3DTC kinase and the CEP-6331 inhibitor, achieving a predicted binding affinity of -12.23 kcal/mol.  
• Analyzed the top binding pose to identify the specific amino acids responsible for the stable interaction, including key hydrophobic (ILE180, LEU274) and polar (LYS228, ASP284) residues.

**Single-Cell Transcriptomic Profiling of the B16 Melanoma's Microenvironment** [GitHub Link](#)  
• Processed raw single-cell transcriptomic data from a B16 melanoma model using R (Seurat), applying quality control, normalization, and clustering to map the tumor's cellular heterogeneity.  
• Characterized the resulting cell clusters by annotating fourteen distinct cell types within the tumor microenvironment and performed differential expression analysis to identify their unique gene signatures.

## Honors and Leadership

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**First runner-up in the Biomimicry Innovation Challenge and Showcase**, Vellore Institute of Technology, 2024

**Top 10 Winner of Bio-Inspired Design Fest (BIDFEST) Ideathon**, Vellore Institute of Technology, 2024

**Vice Chairperson at Alpha Bio Cell, VIT University, India** *January 2024 - December 2024*  
• Led several biotechnology and bioinformatics-related events, including hackathons, workshops, and quizzes, enhancing the club's engagement.  
• Mentored junior core members and applied leadership, technical, and project management skills to ensure successful event planning and execution.

## Extra-Curricular

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**Volunteer | National Service Scheme, NGO**  
*Conducted sessions in government schools on career opportunities and the importance of joining the armed forces.*

**Volunteer | Becoming I Foundation, NGO**  
*Taught science, math, and english in government schools, supporting underprivileged communities.*

**Event Coordinator | BioSummit, School of Biosciences and Technology**  
*Organized a flagship event connecting academia and food industries, managing food distribution and ensuring smooth execution.*